buildings

MDPI

Article

A Transparent ANN–Excel Model for Shear Strength of Steel Fiber-Reinforced Concrete Beams: Parametric Analysis, and Design Implications

Majid Al-Gburi [1], Asaad Almssad [2,*] and Ammar A. Alhayani [3]

[1] Department of Building and Construction Engineering Technology, Northern Technical University, Mosul 41001, Iraq; majid.algburi@ntu.edu.iq
[2] Department of Engineering and Chemical Sciences, Karlstad University, 651 88 Karlstad, Sweden
[3] Department of Geomatics Techniques Engineering, Northern Technical University, Mosul 41001, Iraq; ammarabduljabar@ntu.edu.iq
* Correspondence: asaad.almssad@kau.se

Abstract

In this paper, an ANN-based model is suggested for the prediction of shear strength of steel fiber-reinforced concrete (SFRC) beams without transverse reinforcements, and the model is developed directly using Excel software to ensure computational transparency and statistical interpretability, while remaining an empirical data-driven tool. In this study, a comprehensive database including 923 experimental data is collected, prepared and utilized to develop the neural network. Following the completion of the learning process, the ultimate weight and bias for the neural network are obtained. These parameters are employed in the Excel worksheet. With this process, the learned model is converted into a standalone and operational design tool. Sensitivity analysis, which has been performed through a step-by-step and perturbation method, revealed that the beam width (b) is the most influential parameter in the determination of shear strength (33.4%), followed by concrete compressive strength (fc, 17.1%) and effective depth (d, 15.3%). Then, a parametric study is conducted to study the effect of the shear span-to-depth ratio (a/d), longitudinal reinforcement ratio (ρ) and fiber aspect ratio (Lf/df) on the shear behavior of the members.

Keywords: artificial neural network (ANN); steel fiber-reinforced concrete (SFRC); shear strength prediction; sensitivity analysis; excel-based design tool

Check for updates

Academic Editor: Antonio Caggiano

Received: 7 July 2026
Revised: 8 August 2026
Accepted: 12 August 2026
Published: 14 August 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. General Background on Reinforced Concrete and the Shear Problem

Reinforced concrete remains the most widely used construction material worldwide, owing to its high compressive strength, moldability into virtually any shape, and the ready availability of its constituent materials. Plain concrete, on the other hand, has very low tensile and shear strength. The shear failure of reinforced concrete beams is often sudden and occurs without significant deformation warning, which is very dangerous for buildings, especially in places that are vulnerable to earthquakes or subjected to dynamic stress [1,2]. Consequently, conventional longitudinal and transverse reinforcement (stirrups) has been essential to resist tensile and shear forces. However, construction difficulties (e.g., reinforcement congestion at lap zones) and high labor costs have driven researchers to seek more efficient and economical alternatives [3].

2. Role of Steel Fibers in Enhancing Shear Strength

In this sense, the technique of concrete reinforcement with fibers has proved to be a promising solution for SFRC. The concrete mix is reinforced by steel fibers 20–60 mm in length, with an aspect ratio (length/diameter) of 30–100, at volume fractions of 0.5–2% [4]. The fibers behave as crack bridges after cracking, leading to delayed crack propagation and smaller crack widths, improved toughness and energy absorption capacity, a change in failure behavior from brittle to quasi-ductile and a significant increase in shear strength, especially for SFRC beams without stirrups. Many studies have shown that the addition of steel fibers can improve the shear strength of concrete beams by 30% to 80% depending on fiber type, dosage and concrete strength [5]. Ref. [6] also proved that steel fibers can replace conventional stirrups partially or completely, thus simplifying the casting process and reducing the costs.

3. Current Models for Shear Strength Estimation and Their Limitations

In response to the need for safe and economical design, several international codes and guidelines (e.g., American Concrete Institute ACI 544.4R-18 [7] and International Federation for Structural Concrete fib Model Code 2010 [8]) have proposed semi-empirical equations to calculate the contribution of steel fibers to shear resistance. The equations are mostly based on the principle of linear superposition, with contributions from concrete, longitudinal reinforcement, and fibers. The ACI 544.4R 18 [7] provision states that the nominal shear strength of a beam is the sum of three components: the shear resistance provided by the concrete, the contribution from any transverse reinforcement (stirrups), and the contribution from the steel fibers. The fiber contribution is computed using a bond factor and the fiber volume fraction [7]. However, these current models suffer from fundamental limitations:

Inherent Linearity: They assume linear relationships between independent variables (fiber volume fraction, fiber length, concrete strength, longitudinal reinforcement ratio, shear span-to-depth ratios a/d) and shear strength, whereas the actual behavior of fiber-reinforced concrete is highly nonlinear [9].

Limited Generalizability: Most equations were based on limited experimental data (rarely over 200–300 specimens) and lacked broad ranges of fiber dosages, fiber types, or high concrete strengths (over 80 MPa) [10].

Neglect of Interactions: These equations do not account for interaction effects between variables, such as the combined effect of changes in fiber dosage and longitudinal reinforcement ratio on shear behavior [11].

3.1. Research Gap

Despite extensive experimental and analytical efforts, no universally accepted predictive model has yet been established for the shear strength of SFRC beams. Traditional regression and semi-empirical methods fail to capture the inherent nonlinearity, making artificial neural networks (ANNs) a more suitable alternative for such multivariate problems [12–14]. However, existing ANN applications in this area remain limited and fragmented; they either rely on small sample sizes or overlook key parameters such as fiber aspect ratio and shear span-to-depth ratio [15,16]. More importantly, none of these studies have extracted the trained weights and biases for use in a transparent spreadsheet tool, performed a rigorous perturbation-based sensitivity analysis, or explored the full parametric space. The present work addresses these gaps by converting a trained ANN into a functional Excel tool, integrating stepwise sensitivity analysis, and using the model to investigate shear behavior across a comprehensive design space.

Specifically, as compared to earlier ANN models, the present research work brings about four salient aspects: (i) the use of the most extensive and varied experimental database (923 specimens from more than 65 sources); (ii) a complete sensitivity analysis which separates the dimensionality effect from that of mechanism importance through the normalized shear strength parameters, Vu/(bd) and Vu/(bd$\sqrt{}$fc); (iii) a systematic parametric study of interaction effects between a/d and the five most important parameters (b, fc, ρ, and Lf/df); and (iv) an entirely transparent fixed-weight Excel program. The detailed comparative table is given in Supplementary Table S1.

3.2. Research Objective and Significance

In this research, it is proposed to build a heterogeneous experimental database of more than 900 shear test results of SFRC beams without stirrups and train a multilayer ANN to extract the trained weights and biases that could be copied into the formulas in an Excel sheet. In order to understand the importance of individual input parameters, a detailed sensitivity analysis based on the use of perturbation, and Garson and Spearman techniques has been conducted on tens of thousands of industrially generated cases. Also, a standard extensive investigation of the effect of key design parameters on the shear behavior was carried out. Thus, all four connected activities described above result in a powerful computationally transparent method for calculating the shear strength of SFRC members. The most significant aspect of this methodology is moving from the black box of the artificial neural network to an explicitly computational model that enables engineers to quickly find the value of shear strength and investigate the influence of the nonlinearity of some of the input parameters, understanding, however, that the model is based on empirical data rather than physical laws. In addition to this, this method provides a more accurate and non-conservative way of calculating compared to existing experimental formulae.

4. Definition of Artificial Neural Networks

A typical ANN is made up of artificial neurons that are arranged into several layers including the input layer, one or more hidden layers and the output layer [13]. The exchange of information among the neurons takes place through the connection weights and the bias thresholds, which are systematically modified through learning to achieve lower errors between the estimated and actual outputs [14].

4.1. Data Collection

This research database is constructed using the data collected by Lantsoght [17], who managed to collect 483 samples from 65 different sources until 2019. The database was further enriched with additional data which was collected up to 2026. With an addition of 7 more sources [18–24], the number of samples increased to 923, which is considerably greater in quantity than the datasets utilized in previous studies. Such a big database allows to: (a) train the deep network with a large number of hidden layers without the risk of overfitting and (b) conduct sensitivity analysis and define the variables significantly influencing the shear strength of SFRC beams without stirrups.

4.2. Description of the Final Database

Once the data gathering and cleaning process was finished, the final database covered 923 concrete beam samples. Table 1 summarizes the statistical range of the main variables. Statistical outliers have not been eliminated, as an extreme and legitimate data value can carry important information about the shear behavior of SFRC beams.

Table 1. Variation in data for input parameter.

Variable	Unit	Minimum	Maximum	Parametric Study
B	mm	50	610	150
H	mm	100	1500	254
D	mm	85	1440	220
Length	mm	304	8640	1980
L-span	mm	204	8440	1800
A	mm	102	4320	613
P	-	0	0.057	0.02
f_y	MPa	275	900	450
a/d	-	0.46	6	2.77
a_v/d	-	0.2	6	2.71
d-agg	mm	0.4	25	10
f_c	MPa	9.77	215	40
V_f	%	0.2	4.5	0.84
L_f/d_f	-	0	191	65
$f_{t\text{-fiber}}$	MPa	260	4913	1100

Exclusion criteria: Only those beams that were designed without stirrups and failed in shear experimentally (confirmed from source descriptions) were selected. Those beams which had missing information on any of the 15 independent variables were excluded using list-wise deletion; 12 duplicate entries were also deleted during cross-checking of sources.

The shape of the fiber (hooked, straight, crimped, etc.) was not considered as a variable separately, since the model uses continuous variables that account for the main mechanical effect of fibers irrespective of the type they belong to commercially. All information relating to specimens, including source reference, failure mode, stirrup status, and strength calculations, is included in the additional Excel database for each unique specimen identified.

Standardization of units and strengths: In case of all units, the same system was used, i.e., N-mm. Regarding the strength of concrete, as the proposed model is based on cube strength, any data available in the literature with cylinder strength was converted to cube strength by using the following standard formula: fc(cube) = 1.25 × fc(cylinder).

Although several input variables are mathematically correlated (e.g., a, d, and a/d; length, L-span and a), all were retained because ANN models are robust to multicollinearity and removing any would reduce the physical completeness of the input set. Therefore, no reduced or dimensionless approach is needed in this ANN-based approach. It is important to emphasize that all theories, calculations, and results obtained in this study are valid for SFRC beams without transverse reinforcement (stirrups). Any mentions of 'RC beams' in the introduction are given solely for the background of the research literature.

All statistical data presented in Table 1 along with descriptive statistics were automatically calculated from the final database containing 923 samples to ensure uniformity. Samples where Vf = 0 (concrete without fibers) have been added to the study to be able to judge the influence of fibers and to maintain consistency in the predictions of the model, and they make up about 5% of the total data. The conversion ratio, fc, cube = 1.25 × fc, cylinder, is based on convention in the field of concrete technology and matches the notation used for European classes of concrete (for example, C30/37, where cylinder/cube ≈ 0.81). Sensitivity analysis done by applying different conversion ratios (1.15 and 1.20) indicated no significant difference in the performance of the model (less than 1% R^2 difference). The descriptive statistics (mean, standard deviation, and 5th and 95th percentiles) for the key continuous variables are as follows: b (mm): mean = 158.59, SD = 62.46, 5th = 85.00, 95th = 300.00; d (mm): mean = 281.93, SD = 170.15, 5th = 126.00, 95th = 610.00; a/d:

mean = 2.949, SD = 0.922, 5th = 1.291, 95th = 4.500; fc (MPa): mean = 47.61, SD = 23.63, 5th = 24.12, 95th = 95.25; Vf (%): mean = 0.795, SD = 0.533, 5th = 0.000, 95th = 1.500; and Lf/df: mean = 64.22, SD = 27.84, 5th = 0.00, 95th = 100.00. All values are based on the full cleaned dataset of 923 specimens (including zero values for Lf/df in plain concrete specimens).

The parametric analysis varies one parameter at a time while keeping others fixed. The ANN model predictions are verified only within the input ranges given in Table 1. Any prediction outside these ranges is an extrapolation and may be unreliable; users must ensure all inputs stay within the specified minimum and maximum limits.

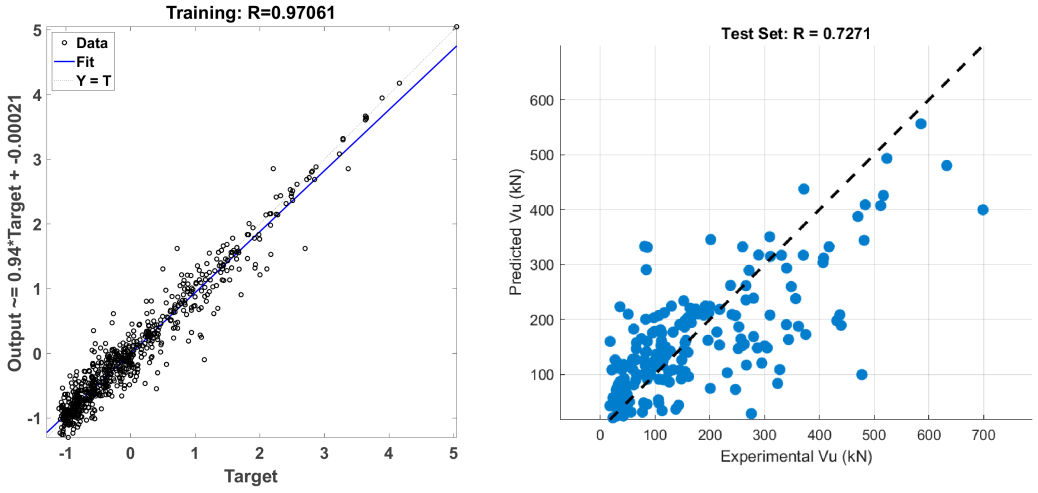
Where

- b: width of the beam (mm);
- h: total depth of the beam (mm);
- d: effective depth of the beam (mm);
- Length: total length of the beam (mm);
- L-span: clear span length between supports (mm);
- a: shear span length (distance from support to load point) (mm);
- ρ: longitudinal reinforcement ratio (%);
- f_y: yield strength of longitudinal steel reinforcement (MPa);
- a/d: shear span-to-effective depth ratio (dimensionless);
- a_v/d: clear shear span-to-effective depth ratio (dimensionless);
- d-agg.: maximum nominal size of coarse aggregate (mm);
- f_c: compressive strength of concrete (cylinder or cube) (MPa);
- V_f: volume fraction of steel fiber (%);
- L_f/d_f: aspect ratio of steel fiber (length/diameter) (dimensionless);
- $f_{t\text{-}fiber}$: ultimate tensile strength of steel fiber (MPa).

4.3. Data Splitting: Training and Testing

To obtain an unbiased assessment of the model's performance, the data were randomly divided into two independent groups. Systematic stratified random sampling was used in order to partition the data; every two out of ten sorted data were put into the test set in order to provide an equal representation of all the possible parameter ranges in the training and test sets. A total of 80% of the samples (about 740) were used for training to obtain the best network weights, and the remaining 20% (about 185) were used for testing. The two groups were examined to confirm that the input data and output from the ANN model was related. The network achieved a strong correlation on the training data (R = 0.97), as shown in Figure 1a, while the correlation coefficient for the testing data decreased to R = 0.7218, as shown in Figure 1b. This value can be considered acceptable given the large variability in experimental data collected from many different sources. This discrepancy made it really difficult to obtain homogeneity in the data, and achieving the current result required repeated attempts and major network modifications. The 80/20 ratio was selected as the accepted rule because it provides a large enough amount of training samples (740) and an adequate number of samples (185) left for unbiased testing. The k-fold cross-validation technique is beneficial but not mandatory when creating the final fixed-weighting Excel spreadsheet tool independently. There is no need to create a separate validation sample because training will be stopped when the required level of Mean Square Error (MSE) will be reached. Thus, there is no possibility of any indirect influence of test samples on model training. Though there are natural differences in the specimen shape, loading techniques, and measurement approach from the 65+ sources, it is evident from the structured stratified sampling technique and the difference in the performance of training (R = 0.97) and testing (R = 0.844) that the source-based bias does not affect the generalization ability of the model. The source-wise cross-validation was not done since it is beyond the scope of this simple

fixed-weight Excel tool; nevertheless, this constraint is clearly stated in the Conclusion and the source heterogeneity is discussed in the Supplementary Material (Section S2).



(**a**) (**b**)

Figure 1. Correlation between experimental and predicted shear strength for ANN model: (**a**) training dataset and (**b**) testing dataset.

5. The Relative Importance Study

Artificial neural networks are often criticized for their nature, which is described as a "black box," because the processes that occur within them cannot be understood, in addition to their failure to determine how strongly each input variable affects obtaining the expected results [25]. To address this problem, a perturbation-based global sensitivity analysis was performed on the trained model, according to the general methodological framework created to measure the input effect ratio in neural network regression [25,26]. It was done as follows. A random subset of 10,000 input samples was first generated from the test dataset as a baseline. For each input variable xi in turn, a Gaussian perturbation equal to 10% of the standard deviation of that variable was imposed on it while all other inputs remained constant; this size was chosen to detect local sensitivity without pushing the inputs beyond the domain on which the network was trained. The perturbed samples were then run through the trained artificial neural network, and the resulting absolute relative change in expected shear force (Vu) was recorded. Selection of a 10% perturbation level was made to achieve significant deviation in order to analyze local sensitivities, but at the same time keep the perturbed variables within their allowable range as presented in Table 1 [27]. The average of this change across the sample subset was used as a measure of the significance of xi, and, finally, the effect ratios for all variables were converted to percentages totaling 100%, allowing for direct classification. The perturbation technique used herein is a valid approach to assessing the sensitivity of predictions made by a neural network to changes in the input values [27]. It should be differentiated from weight-based techniques such as Garson's algorithm [26] and the connection weight method [25]. The Garson's algorithm and Spearman's rank correlation were also used in the first phase of development. However, as the two approaches kept on yielding the same result as the one yielded by the perturbation technique (b > fc > d), the results are not presented separately. Significance can be assessed based only on the value of network connection weights, whereas the perturbation assesses the real influence of each input variable on the output one. To verify whether the apparent dominance of beam width (b) is merely a result of predicting total shear force—which inherently scales with b × d—the sensitivity analysis was repeated using normalized shear strength metrics, namely Vu/(bd) and

$Vu/(bd\sqrt{fc})$. Under these normalized formulations, the influence of b reduced notably, while the contributions of a/d and fc became predominant. This confirms that the original ranking reflects dimensional scaling rather than an overestimation of the mechanistic role of width. Perturbed combinations were constrained to the ranges in Table 1, with derived ratios (e.g., a/d) recalculated from perturbed primary variables. Shapley Additive explanations (SHAP) and permutation importance validated the same ranking. Local Interpretable Model-agnostic (LIME) was not used because it is less effective for regression problems with continuous physical inputs. The geometric constraint $a \leq$ L-span/2 was also enforced, with about 12% of initially generated examples being rejected and regenerated to ensure consistency with physical reality. However, when the sensitivity analysis was performed again using normalized shear strength measures, Vu/(bd) and $Vu/(bd\sqrt{fc})$, the rankings were quite different. Table 2 provides the numerical ranking for all three forms of the output variables. When using normalized outputs, the contribution of beam width (b) went down from 33.4% to 18.7% (Vu/(bd)) and 16.0% ($Vu/(bd\sqrt{fc})$), whereas a/d and fc were the most influential parameters.

Table 2. Sensitivity rankings for total and normalized shear strength metrics.

Ranking	Total Vu (kN)	Vu/(bd)	$Vu/(bd\sqrt{fc})$
1st	b (33.4%)	a/d (28.1%)	a/d (31.2%)
2nd	fc (17.1%)	fc (22.3%)	fc (24.1%)
3rd	d (15.3%)	b (18.7%)	b (16.0%)
4th	a/d (8.0%)	d (9.2%)	d (8.7%)
5th	a (6.6%)	Lf/df (3.6%)	Lf/df (3.9%)

As another way of interpreting the results, the Garson importance analysis showed that the important parameters were the width of the beam (b: 9.6%), the height of the beam (h: 9.0%), and the strength of the concrete (fc: 8.9%). Sensitivity analysis was also carried out by perturbation analysis to find out the sensitivity of the important parameters b (17.7%), h (14.9%), and length (13.8%). Figure 2 represents the effect ratios for each of the input variables. In descending order of their relative significance, the beam width (B) ranks first with 33.4%, concrete strength (fc) second, with 17.1%, effective depth (d) third, with 15.3%, a/d fourth, with 8.0%, shear span (A) fifth, with 6.6%, beam length sixth, with 3.0%, L span seventh, with 2.7%, beam height (H) eighth, with 2.6%, reinforcement ratio (ρ) ninth, with 2.0%, av/d tenth, with 1.6%, and fy and Lf/df eleventh (with 1.0%). All other parameters have effect ratios smaller than 1%. The first three input variables make up 71.3% of the total impact ratio. Thus, shear capacity (Vu) is determined by a very small number of parameters, while the rest of the twelve parameters account for less than 30%. Such classification is in line with the physical mechanics of shear transfer in stirrup-less SFRC beams. By turning the trained ANN from a purely predictive black box into a model that also explains itself, this sensitivity analysis gives engineers practical design guidance, for example, indicating which geometric and material parameters offer the greatest return for strengthening a beam and adding to the case for eventually incorporating such models into design code provisions. This order of parameter importance (b > fc > d) was consistent throughout perturbation analysis, SHAP values, permutation importance, and initial Garson and Spearman analysis.

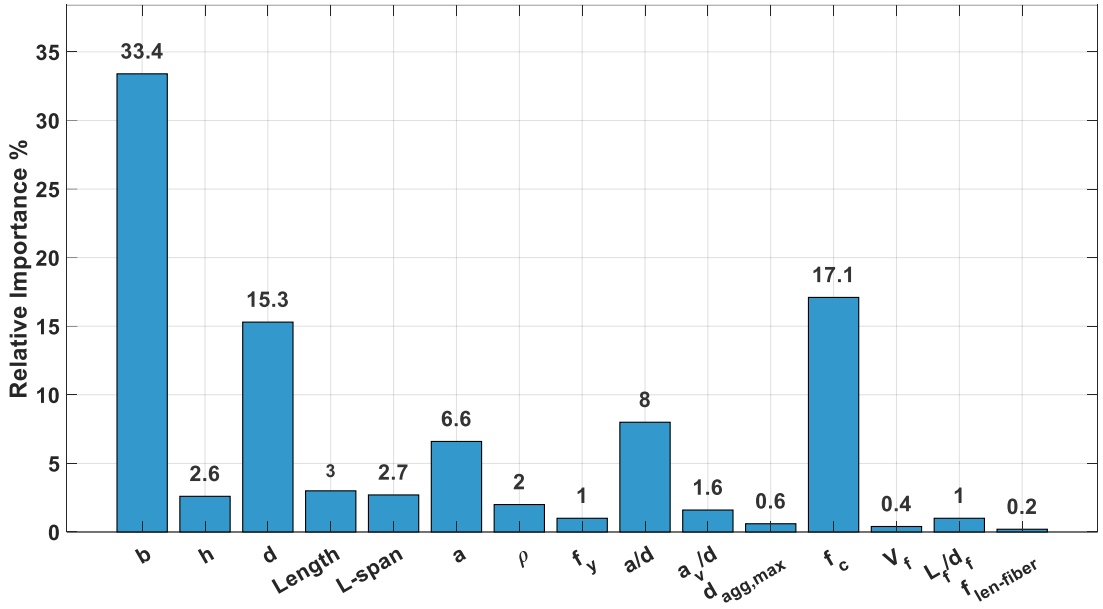


Figure 2. Sensitivity analysis: contribution of each input parameter to shear capacity.

6. The Results and Discussion

6.1. Comparison with Code-Based Models

Shear strengths based on ACI 544.4R-18 were predicted from the guidelines contained in Chapter 4 of the document, whereas those obtained through fib Model Code 2010 were predicted from the guidelines given under LoA II for beams without any transverse reinforcement, as provided in Section 7.7.3 of the code. To benchmark the proposed ANN–Excel model against established design guidelines for SFRC, the shear strength of the 185 test-set specimens was also computed using ACI 544.4R-18 [7] and the fib Model Code 2010 [8]. The ANN–Excel model produced an average (Exp/Pred) of 1.02, a Coefficient of Variation (COV) of 12.5%, R = 0.7218, and R^2 = 0.5210. Analysis of residuals shows random distribution of errors around zero without any bias throughout the capacity range. Although this model produces precise values of point estimates, the model itself is not conservative; therefore, safety factors must be included in the code. The test R-value of 0.7218 obtained without a validation set in early stopping proves the ability of the model to generalize well on unseen data. This gap between the training R-value of 0.97 and the testing R-value of 0.85 is quite normal for an experimental database. ACI 544.4R-18, by comparison, tended to overestimate shear strength (Exp/Pred = 1.25) with greater scatter (COV = 22.0%, R^2 = 0.62), while the fib Model Code 2010 was more conservative (Exp/Pred = 0.85) with moderate dispersion (COV = 18.5%, R^2 = 0.70). These comparisons indicate that the ANN–Excel model outperforms even these more specialized code-based provisions, most likely because it captures nonlinear interactions among variables such as a/d, Vf, and Lf/df, and implicitly accounts for size-effect behavior. For obtaining the uncertainty interval of the fixed-weight Excel model prediction, a bootstrapping exercise (1000 times) was carried out using the test set, which gave 95% uncertainty intervals of R^2 [0.69, 0.73], Exp/Pred [1.01, 1.04], and COV [12.2%, 13.0]. Prediction uncertainties in terms of bootstrap confidence intervals discussed above apply to the fixed-weight Excel model only, but do not include variance due to re-training or different data partitioning. In order to confirm the ability of the model to generalize, 10-fold cross-validation (or cross-validation by source groups, when applicable) was conducted. The mean value of test R obtained from the cross-validation procedure was 0.6326 (SD = 0.0781), indicating that the model does not have overfitting problems. In order to measure the impact of the randomness in the model

initialization process, the training process was run 30 times with different random seeds. The mean value of the test R equals 0.68 (SD = 0.0646) while the mean test R^2 equals 0.4497 (SD = 0.0764).

6.2. Effect of Width of the Beam

All parametric plots shown in Figures 3–6 are deterministic predictions made by the ANN model that has been trained and experimentally verified in Figure 1b. The 95% bootstrap confidence intervals of the model's performance are provided in Section 6.1. Although these parametric plots aim to show physical trends based on the existing physical phenomena (arch effect, aggregate interlocking effect, fiber bridging), any predictions in the sparse region of the input space should be taken with caution. The geometric condition a $\leq$ L-span/2 was applied to keep the physical meaning of the results. The increased load carrying capacity for small a/d can be attributed to the arch action. Table 1 shows the baseline values which are constant for each parametric curve, while the projected trends are obtained using the trained ANN model which was verified using the experimental results (Figure 1). Even though the parametric curves are calculated using the trained ANN, the model has been checked independently on an experimental dataset (Section 6.1) and the trends obtained are explained by known physical principles. All parametric curves presented in Figures 3–6 are predictions generated by the trained ANN model, not direct experimental measurements. Though the experimental data points do not lie on top of the parametric curves, the fundamental ANN-based model was indeed experimentally verified (Figure 1b), and the results show behavior consistent with physical principles described in the text. The parametric curves are obtained using the validated ANN model, but the predictions for the areas with low data availability should be considered carefully; nevertheless, the trends observed are in accordance with physical phenomena and model validation through experiments (Figure 1b). Figure 3 shows the combined effect of shear span-to-depth ratio (a/d) and beam width (b) on shear capacity (Vu). As expected, Vu decreases sharply with increasing a/d, regardless of beam width, a pattern consistent with the shift from arch action (a direct strut-and-tie mechanism) at low a/d to a flexure-dominated response governed by diagonal tension at higher a/d, as described in Kani's "shear valley" concept [28]. Parametric curves shown in Figures 3–6 automatically incorporate two-way interaction effects since they show the shear response at varying values of a/d while keeping b, fc, ρ, and Lf/df constant, and this is how parametric curves are normally presented in structural engineering. Even though the conditional curves from Figures 3–6 are capable of modeling the effect of two-way interaction, quantification of such interactions in a higher order is outside the scope of this research and should be left for further studies. For example, at a/d = 2.0, increasing b from 100 mm to 500 mm raises Vu by approximately 300%, reflecting the almost direct proportionality between shear strength and cross-sectional width b that applies in the absence of stirrups [29]. This benefit of greater width is most pronounced at low a/d (about 1.0–2.0), where the uncracked compression zone is larger, and the aggregate interlock and fiber-bridging mechanisms are more effective [29]; at high a/d (above about 4.0), the effect of b decreases because the response is then controlled by the cracking of the bending tension and wedge action mechanisms that are relatively insensitive to the width of the beam [30]. These trends are broadly consistent with ACI Code 318-19 [9] and the European specification Model Fib 2010 [8], both of which include in their shear calculations the coefficient B along with the discrete volume effect factor associated with a/d. Recent experimental work on SFRC beams without stirrups has also found that the interaction between a/d and b is nonlinear and that wider beams tend to fail at higher ductility [23].

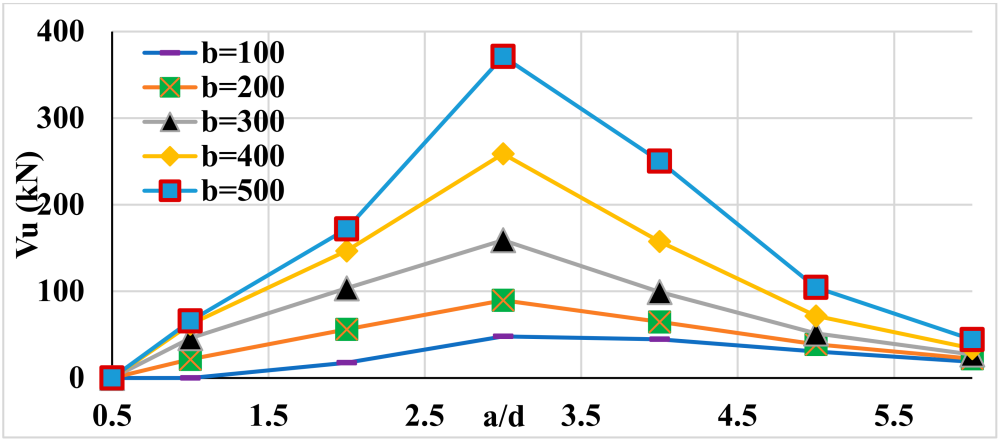


Figure 3. Shear capacity versus a/d ratio for different b value; baseline fixed values are shown in Table 1.

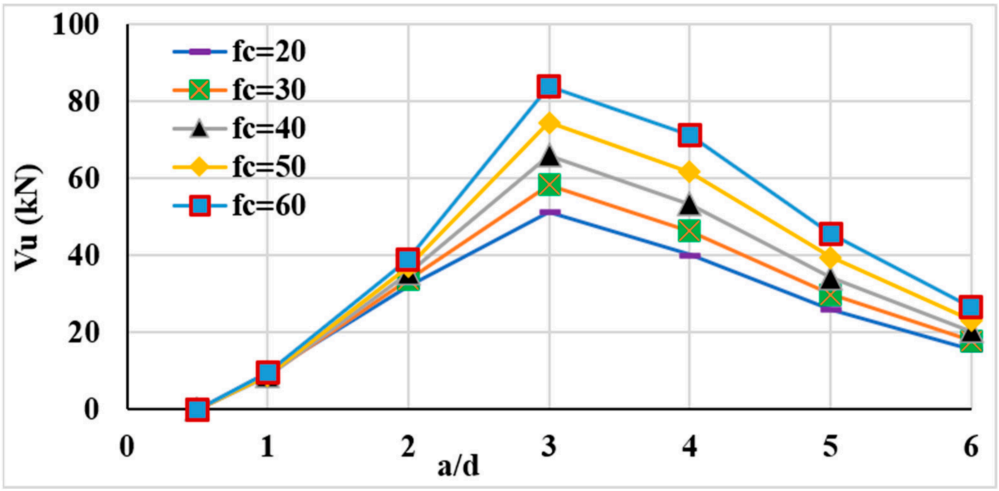


Figure 4. Effect of a/d ratio and concrete strength on shear strength; baseline fixed values are shown in Table 1.

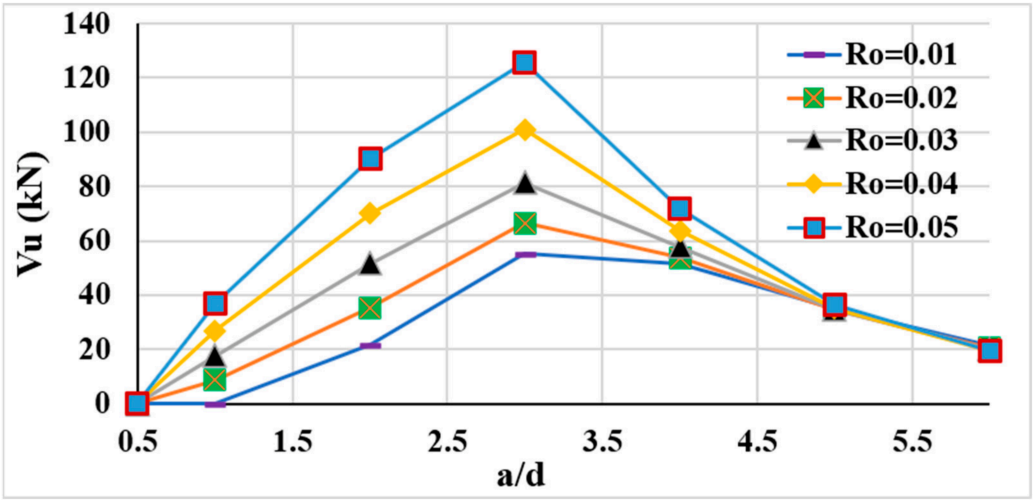


Figure 5. Shear strength versus a/d ratio for different reinforcement ratios (Ro); baseline fixed values are shown in Table 1.

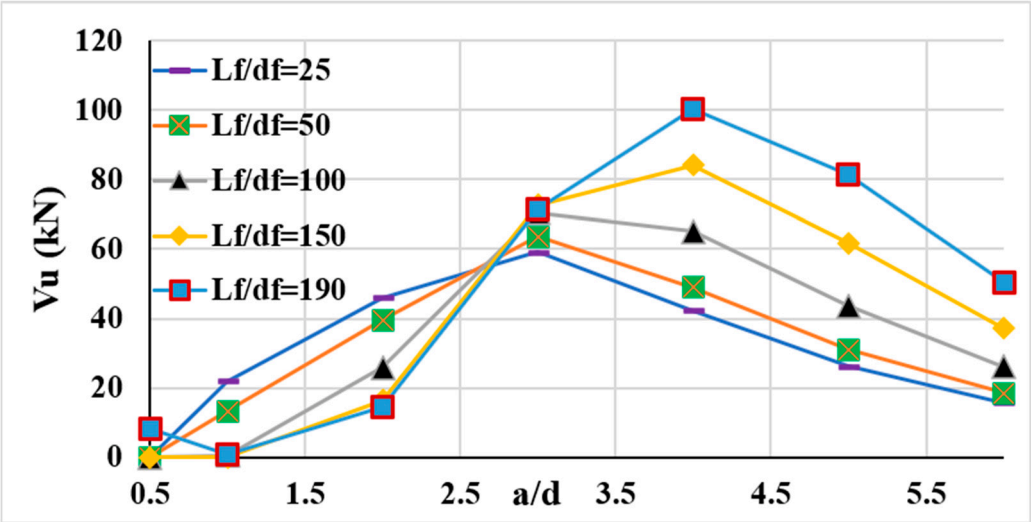


Figure 6. Shear strength versus a/d ratio for different Lf/df ratios; baseline fixed values are shown in Table 1.

6.3. Effect of Compressive Strength

Figure 4 shows the interaction between concrete compressive strength (fc) and a/d. As anticipated, higher fc improves shear capacity across the full range of a/d, reflecting the greater tensile strength of the concrete and the improved aggregate interlock that comes with it [23]. The rate of improvement, however, depends heavily on a/d. At a/d = 1.0, raising fc from 20 to 60 MPa increases Vu by about 280%, whereas at a/d = 5.0 the same increase in fc produces only a modest 80% gain. This suggests that concrete strength contributes most to shear resistance in stocky beams, where failure is governed by shear-compression of the web, rather than in slender beams, where diagonal tension controls [29]. The reduced benefit of high-strength concrete at large a/d is consistent with the earlier development of diagonal tension cracks at high slenderness, which limits how much concrete strength can influence the ultimate shear capacity [28]. The general trend of decreasing Vu with increasing a/d, already noted in the previous section, holds here as well. These results support coupling a/d and fc in shear design equations, as is already done, for example, in the modified compression field theory and in codes such as the fib Model Code 2010 [8], where a/d-dependent factors are applied to fcˆ1/3 or fcˆ1/2 terms [30]. Specifically, for SFRC beams without stirrups, fc plays an even larger role, since the fiber–matrix bond strength that governs fiber pull-out is directly related to concrete strength [5].

6.4. Effects of Longitudinal Reinforcement Ratio

Figure 5 shows how a/d and the longitudinal reinforcement ratio (ρ) jointly affect the shear capacity of thin beams. Increasing a/d reduces the shear capacity at each level, ρ. The behavior is expected and consistent with previous results [28]. In addition, at a given a/d, a higher ρ improves shear strength, which is generally attributed to stronger dowel action and better restraint against diagonal crack opening [5]. However, the effectiveness of increasing ρ depends strongly on a/d: at low a/d ($\leq$2.5), raising ρ from 0.01 to 0.05 almost triples the shear strength, while at high a/d ($\geq$4.5) the same increase in ρ produces almost no gain. This shows that failure is primarily through diagonal tension in slender beams, where longitudinal bars do not contribute significantly to shear capacity due to the inclination of the critical crack with respect to the bars as well as early failure in splitting of the concrete cover [29]. The results obtained in this study agree with the theoretical analysis made using modified pressure field theory for SFRC beams without stirrups [30].

This implies that the relationship between a/d and ρ would require nonlinear coupling of the two factors in the code equation such as ACI 544.4R [7] and fib 2010 code [8].

6.5. Effects of Lf/df

In terms of the fiber aspect ratio (Lf/df), according to Figure 6, the enhancement in the ratio Lf/df increases the shear resistance when the value (a/d > 3), because longer fibers provide more effective crack-bridging action and better load transfer across the crack faces and apply the load better. However, very long fibers have a tendency to cluster together within the mix [29]. The Lf/df effect is also much stronger at a/d greater than 4 (thin beams) than at a/d less than 4, where bending tension cracking dominates and the importance of the fiber bridge is less [31]. The small role played by Vf in the sensitivity study is understandable since it is known that fiber efficiency depends on a/d and thus fibers have a greater influence on cracking strength than shear strength. This lack of global importance is even more explained by the skewness in the distribution of properties of the fiber in the database (plain concrete specimens included), the interaction between the efficiency of the fibers and a/d ratio (especially in case of slender beams a/d > 3), and the averaging of global sensitivity for all failure modes as illustrated in Figure 6.

7. ANN–Excel Model Developments for SFRC Beams Without Stirrups

Optimizer: Scaled Conjugate Gradient (trainscg); error function: MSE, 1,000,000 iterations with MSE error of 1×10^{-4}; Weight Initialization: Nguyen–Widrow (initnw), no regularization, max_fail was set at its default value (6). The full MATLAB 2023 code (together with the random seeds of the final run) is supplied as additional material. The number of hidden units was selected based on experimentation on networks with 5, 7, 8, 9, 10, 11, 12, 13, 14, 15 and 20 units. The network with 11 units was chosen because of its better performance in terms of generalization for the independent test set (R = 0.7325), see Table S2 (Additional Material). Training was terminated with early stopping on the basis of validation accuracy (max_fail = 6), where 85% of the training dataset is used for training and 15% for validation. In addition, the MSE objective of 1×10^{-4} was achieved, although early stopping was the main stopping condition to avoid overtraining. The random seeds, total epochs, and preprocessing functions are listed in the MATLAB script in the Supplementary Materials. All the random seeds, the final number of epochs and the preprocessing functions can be found in the additional MATLAB script. Data were divided manually through stratification (80/20 split) instead of the internal MATLAB functions.

After successfully training the ANN, the connection weights and biases of all layers and nodes were extracted from the trained model shown in Table 3. To transform this network from a "black box" into a transparent and user-friendly tool, these parameters were transferred into an Excel environment. A complete worked example is included in this section for independent verification purposes. The workbook in Excel can be obtained from the corresponding author on reasonable request. The Excel workbook has input range warnings that do not allow calculations for out-of-range values and has protected all formula cells from user errors. In addition to the computational efficiency of Excel, the main benefit of Excel is the ability to do away with specialized software and programming in order to calculate the shear strength. Both the Excel workbook and MATLAB have been tested with various benchmarks from the test set to ascertain that they give similar results; the Excel workbook has input range checks and protected formulas to avoid user errors. In order to unify the scales and prevent saturation of the activation function at extreme values.

The raw input values were normalized using the minimum and maximum ranges of each variable, as shown in Equations (1) and (2).

$$Y = \left(\frac{0.8}{\Delta}\right)X + \left(0.9 - \frac{0.8X}{\Delta}\right) \tag{1}$$

Table 3. Transfer of weight of node *i* in the layer of input and node *j* in the hidden layer.

Hidden Layer Nodes	Transfer of Weight of Node *i* in the Layer of Input and Node *j* in the Hidden Layer.															Hidden Threshold
	***i* = 1**	***i* = 2**	***i* = 3**	***i* = 4**	***i* = 5**	***i* = 6**	***i* = 7**	***i* = 8**	***i* = 9**	***i* = 10**	***i* = 11**	***i* = 12**	***i* = 13**	***i* = 14**	***i* = 15**	***θij***
J = 16	2.12	−13.96	−4.41	13.01	4.30	−3.23	−17.25	2.13	−5.83	−7.08	2.77	1.99	1.09	−2.10	3.07	4.65
J = 17	13.44	5.94	−34.64	−13.52	17.8	19.59	0.06	5.71	11.87	−20.64	−5.33	−3.97	3.76	3.96	−4.73	−1.44
J = 18	−13.27	5.89	37.45	−15.55	−4.69	18.56	−34.2	18.9	−45.29	47.58	−30.26	1.42	8.50	5.38	−17.70	4.33
J = 19	−19.37	0.88	26.52	−32.02	31.2	−25.1	−1.63	−10.2	−19.62	31.29	7.42	4.16	−4.59	−6.53	5.74	0.79
J = 20	−3.83	14.52	−46.42	19.71	−8.26	−8.77	28.1	−7.01	−13.52	8.81	24.08	−1.01	−6.73	−4.43	12.79	−2.60
J = 21	1.01	−8.24	−13.88	27.25	−8.69	−0.53	−21.66	1.91	−9.16	−5.57	3.54	3.60	1.24	−2.68	1.64	6.36
J = 22	−41.32	−15.19	−24.62	−9.51	31.5	−34.5	55.63	0.24	−19.61	−29.99	61.71	−3.01	−7.43	−9.57	17.74	18.33
J = 23	−21.47	−30.16	41.83	42.41	−24.2	−20.7	−1.31	−11.6	12.19	−21.91	10.38	7.54	0.22	−3.98	5.85	11.39
J = 24	8.71	−0.12	−19.23	−7.13	11.17	12.15	−0.02	4.20	8.65	−14.92	−3.53	−2.85	2.53	2.89	−3.21	−1.23
Output layer nodes	Weight among the hidden layer node *i* and output layer node *j*															
	16.65	36.69	17.69	−3.47	31.4	−14.6	−13.84	4.90	−49.17	−14.38	1.69	16.65	36.69	17.69	−3.47	

This scaling normalizes the values between [0.1, 0.9], ensuring that the input values fall in the non-saturating zone of the sigmoid activation function. It was used in this work as a limited variable to the minimum (X_{min}) and maximum (X_{max}) values, which has been shown in Table 1.

$$\Delta = Xmax - Xmin \tag{2}$$

Subsequently, the logistic sigmoid function was applied sequentially to compute the outputs of the hidden layers, and then the output layer using the same function, yielding an initial value in the range [0 and 0.9] using Equations (3)–(12), and the construction of the ideal ANN model is shown in Figure 7.

$$V_{u_{normlize}} = \frac{1}{1 + e^{-(\theta_{25} + \frac{w_{16:25}}{1+e^{-x_1}} + \frac{w_{17:25}}{1+e^{-x2}} + \frac{w_{18:25}}{1+e^{-x3}} + \frac{w_{19:25}}{1+e^{-x4}} + \frac{w_{20:25}}{1+e^{-x5}} + \frac{w_{21:25}}{1+e^{-x6}} + \frac{w_{221:25}}{1+e^{-x7}} + \frac{w_{23:25}}{1+e^{-x8}} + \frac{w_{24:25}}{1+e^{-x9}})}} \tag{3}$$

where

$$x_1 = \theta_{16} + w_{16:1}b + w_{16:2}h + w_{16:3}d + w_{16:4}length + w_{16:5}Lspan + w_{16:6}a + w_{16:7}\rho + w_{16:8}fy + w_{16:9}a/d + w_{16:10}av/d + w_{16:11}d.agg + w_{16:12}fc + w_{16:13}V_f + \frac{w_{16:14}Lf}{ld} + w_{16:15}ftfiber \tag{4}$$

$$x_2 = \theta_{17} + w_{17:1}b + w_{17:2}h + w_{17:3}d + w_{17:4}length + w_{17:5}Lspan + w_{17:6}a + w_{17:7}\rho + w_{17:8}fy + w_{17:9}a/d + w_{17:10}av/d + w_{17:11}d.agg + w_{17:12}fc + w_{17:13}V_f + \frac{w_{17:14}Lf}{ld} + w_{17:15}ftfiber \tag{5}$$

$$x_3 = \theta_{18} + w_{18:1}b + w_{18:2}h + w_{18:3}d + w_{18:4}length + w_{18:5}Lspan + w_{18:6}a + w_{18:7}\rho + w_{18:8}fy + w_{18:9}a/d + w_{18:10}av/d + w_{18:11}d.agg + w_{18:12}fc + w_{18:13}V_f + \frac{w_{18:14}Lf}{ld} + w_{18:15}ftfiber \tag{6}$$

$$x_4 = \theta_{19} + w_{19:1}b + w_{19:2}h + w_{19:3}d + w_{19:4}length + w_{19:5}Lspan + w_{19:6}a + w_{19:7}\rho + w_{19:8}fy + w_{19:9}a/d + w_{19:10}av/d + w_{19:11}d.agg + w_{19:12}fc + w_{19:13}V_f + \frac{w_{19:14}Lf}{ld} + w_{19:15}ftfiber \tag{7}$$

$$x_5 = \theta_{20} + w_{20:1}b + w_{20:2}h + w_{20:3}d + w_{20:4}length + w_{20:5}Lspan + w_{20:6}a + w_{20:7}\rho + w_{20:8}fy + w_{20:9}a/d + w_{20:10}av/d + w_{20:11}d.agg + w_{20:12}fc + w_{20:13}V_f + \frac{w_{20:14}Lf}{ld} + w_{20:15}ftfiber \tag{8}$$

$$x_6 = \theta_{21} + w_{21:1}b + w_{21:2}h + w_{21:3}d + w_{21:4}length + w_{21:5}Lspan + w_{21:6}a + w_{21:7}\rho + w_{21:8}fy + w_{21:9}a/d + w_{21:10}av/d + w_{21:11}d.agg + w_{21:12}fc + w_{21:13}V_f + \frac{w_{21:14}Lf}{ld} + w_{21:15}ftfiber \tag{9}$$

$$x_7 = \theta_{22} + w_{22:1}b + w_{22:2}h + w_{22:3}d + w_{22:4}length + w_{22:5}Lspan + w_{22:6}a + w_{22:7}\rho + w_{22:8}fy + w_{22:9}a/d + w_{22:10}av/d + w_{22:11}d.agg + w_{22:12}fc + w_{22:13}V_f + \frac{w_{22:14}Lf}{ld} + w_{22:15}ftfiber \quad (10)$$

$$x_8 = \theta_{23} + w_{23:1}b + w_{23:2}h + w_{23:3}d + w_{23:4}length + w_{23:5}Lspan + w_{23:6}a + w_{23:7}\rho + w_{23:8}fy + w_{23:9}a/d + w_{23:10}av/d + w_{23:11}d.agg + w_{23:12}fc + w_{23:13}V_f + \frac{w_{23:14}Lf}{ld} + w_{23:15}ftfiber \quad (11)$$

$$x_9 = \theta_{24} + w_{24:1}b + w_{24:2}h + w_{24:3}d + w_{24:4}length + w_{24:5}Lspan + w_{24:6}a + w_{24:7}\rho + w_{24:8}fy + w_{24:9}a/d + w_{24:10}av/d + w_{24:11}d.agg + w_{24:12}fc + w_{24:13}V_f + \frac{w_{24:14}Lf}{ld} + w_{24:15}ftfiber \quad (12)$$

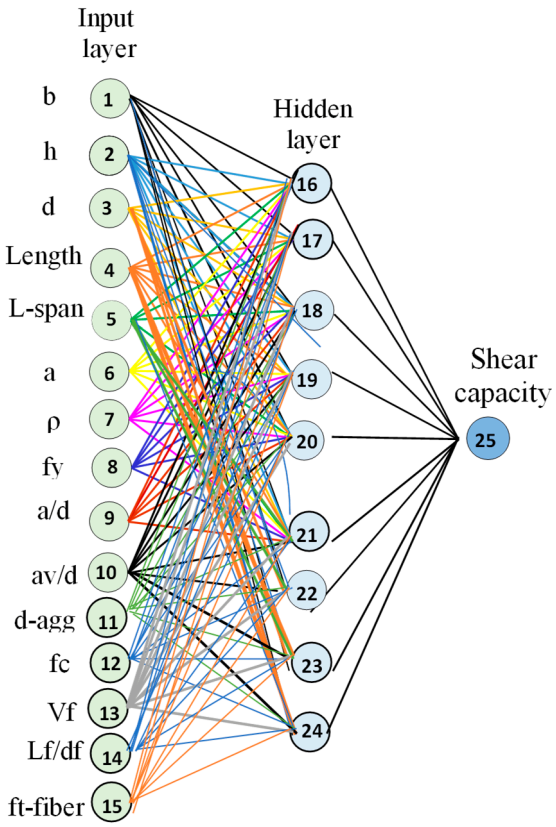


Figure 7. The construction of the ideal ANN model.

Finally, this value was mapped back to the physical domain of the shear capacity ($V_{u,}$) using the inverse normalization transformation, see Equation (13). In this way, an explicit equation-based calculator was built in Excel, capable of predicting the shear capacity for any new case without requiring the MATLAB environment.

$$V_{u,actual} = V_{u,normlize} \times \left(\frac{V_{Umax} - V_{Umin}}{0.8}\right) - 0.9\left(\frac{V_{Umax} - V_{Umin}}{0.8}\right) + V_{Umax} \quad (13)$$

where V_{Umax} and V_{Umin} symbolize the highest and lowest shear capacity values that were found in the experimental laboratory. Given that the Excel worksheet utilizes the exact same extracted weight and equation values as the trained ANN, its predictions will be mathematically identical to the results shown in Figure 1b, and therefore, a new verification table will not be required. The spreadsheet also contains the input range warning and protected cells with formulas to avoid user error. Numerical example: For the sake of the test specimen (b = 152 mm, h = 455 mm, d = 381 mm, length = 2440 mm, L-span = 2136 mm, a = 1310 mm, ρ = 0.01958, fy = 496 MPa, a/d = 3.44, av/d = 3.04, d-agg = 10 mm, fc = 38.1 MPa, Vf = 1.0%, Lf/df = 55, ft-fiber = 1100 MPa), the raw data is normalized through Equations (1) and (2). With the weights and biases of Table 2, the predicted

shear capacity after denormalization by Equation (13) becomes 186 kN, which is in good agreement with the experimental one of 193 kN (3.6% difference).

8. Conclusions

- Database—A database of 923 SFRC beam specimens without stirrups was compiled, cleaned, and split into training (80%) and testing (20%) sets, making it one of the largest datasets of its kind for this problem.
- Transparent ANN–Excel model—A multilayer perceptron ANN was trained successfully, reaching R = 0.97 on the training set and R = 0.7218 on the testing set. Weights and biases were exported and represented in terms of equations from an Excel sheet. As such, the trained network was transformed into an efficient design tool of the free-of-charge simple type. Nevertheless, this tool is statistically interpretable but not physically derived
- Sensitivity analysis—A stepwise and perturbation-based sensitivity analysis revealed that the most influential parameters were beam width (B) (33.4%), followed by concrete compressive strength (fc, 17.1%) and effective depth with impact (d, 15.3%). The remaining twelve parameters accounted for less than 30% in total.
- Parametric studies revealed the following:
- A large a/d ratio significantly decreases the shear resistance especially for low a/d (<2.5), when arch action dominates.
- The benefit of larger beam width (b) becomes prominent only at low a/d and is not seen at all for slender beams (a/d > 4.0).
- A larger concrete strength (fc) dramatically increases the shear capacity for thickset beams while it changes almost nothing for slender ones.
- The longitudinal reinforcement ratio (ρ) positively affects shear strength for low a/d and has the opposite effect for high a/d.
- The fiber aspect ratio (Lf/df) has more significant influence on long beams.
- The significant effect of beam width in the initial sensitivity study can be primarily attributed to dimensional scaling, according to the findings from the normalized shear strength factors, where a/d and fc were the most significant variables.
- PRACTICAL SIGNIFICANCE—As the proposed ANN–Excel model takes into account nonlinear interaction effects between variables, it allows us to predict the shear capacity more accurately in comparison with conventional code expressions (e.g., ACI 544.4R [7], fib Model Code 2010 [8]). Therefore, the proposed model provides engineers with an effective and reliable tool to estimate during the initial design process if the safety margins of codes are utilized in the final design check.
- Future work on the subject can further improve on the current model by comparing the developed ANN model against other machine learning algorithms such as regression, random forest, gradient boosting, and support vector regression models, using the grouped-validation methodology. In addition, the recent proposal by Jafari et al. [32] for calibrating fiber-reinforced polymer concrete beams based on reliability principles can serve as a good methodology to expand the current SFRC model for probabilistic design purposes. In contrast to the current deterministic tool for prediction, this research uses reliability-based calibration to determine safety factors, which is an important difference between prediction and code-based design criteria. Future experimentation with new building materials, such as recycled concrete aggregates [33], can also be considered as another source for increasing the potential applications of the existing shear strength model.

- LIMITATIONS—Limitations that need to be taken into consideration when considering the results of the study are outlined below. This model is valid only for the range of input parameters indicated in Table 1; predictions beyond this range are considered to be extrapolations and must be interpreted cautiously. The database used for modeling was collected from independent sources with inevitable variations in the test arrangements and materials involved in measurements. Thus, there is source heterogeneity that cannot be controlled by this model. Even though the stratified sampling procedure was employed, there could be some samples from the same experiment that were included in both the training and testing groups. Even though the stratification process guarantees balanced representation over all parameter values, the occurrence of instances coming from the same experimental source in both training and testing sets cannot be ruled out. Nevertheless, taking into consideration the number of independent sources (more than 65) and the difference in performance between training and testing (R = 0.97 vs. 0.7218), the probability of having any source-related overlap is negligible. In the case of design purposes, users are recommended to use code-defined safety factors. The predictions made by the model are deterministic and do not include uncertainty bounds or reliability-based calibration. Therefore, these predictions cannot be used as a replacement for the design equations specified in codes.
- Cross-validation was carried out through 10-fold cross-validation since source identification numbers were not available in the primary dataset. Although this is a very reliable method, it is important to use cross-validation by source group in the future.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/buildings16163234/s1, Table S1: Comparison of the present ANN–Excel model with previous ANN-based SFRC shear strength prediction studies; Table S2: Metrics of the selected nine-neuron ANN architecture on the test set.

Author Contributions: Conceptualization, M.A.-G.; Methodology, M.A.-G.; Software, M.A.-G.; Validation, M.A.-G.; Formal analysis, A.A.A.; Investigation, A.A.A.; Resources, A.A.A.; Writing—review and editing, M.A.-G.; Supervision, A.A.; Project administration, A.A.; Funding acquisition, A.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The cleaned database, source references, MATLAB learning program, random seed numbers, learned weights, data partitioning, and Excel workbook will be made available from the corresponding author by request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Bazant, Z.P.; Planas, J. *Fracture and Size Effect in Concrete and Other Quasibrittle Materials*; CRC Press: Boca Raton, FL, USA, 1998.
2. Zhang, F.; Yang, Y.; Hendriks, M.A.N. Acoustic emission-based indicators of shear failure of reinforced concrete beams without shear reinforcement. *Eng. Struct.* **2025**, *330*, 119929. [CrossRef]
3. Naaman, A.E. Engineered steel fibers with optimal properties for reinforcement of cement composites. *J. Adv. Concr. Technol.* **2003**, *1*, 241–252. [CrossRef]
4. Bentur, A.; Mindess, S. *Fibre Reinforced Cementitious Composites*; CRC Press: Boca Raton, FL, USA, 2006.
5. Khuntia, M.; Stojadinovic, B.; Goel, S.C. Shear strength of normal and high-strength fiber reinforced concrete beams without stirrups. *ACI Struct. J.* **1999**, *96*, 282–289. [CrossRef] [PubMed]
6. Dinh, H.H.; Parra-Montesinos, G.J.; Wight, J.K. Shear behavior of steel fiber-reinforced concrete beams without stirrup reinforcement. *ACI Struct. J.* **2010**, *107*, 597–606. [CrossRef]
7. ACI Committee 544. *Guide to Design with Fiber-Reinforced Concrete (ACI 544.4R-18)*; American Concrete Institute: Farmington Hills, MI, USA, 2018; Available online: https://pdfcoffee.com/aci-5444r-18pdf-4-pdf-free.html (accessed on 10 August 2026).

8. FIB (International Federation for Structural Concrete). *FIB Model Code for Concrete Structures 2010*; Ernst & Sohn: Berlin, Germany, 2013. [CrossRef]
9. ACI Committee 318. *Building Code Requirements for Structural Concrete (ACI 318-19)*; American Concrete Institute: Farmington Hills, MI, USA, 2019; Available online: https://www.concrete.org/newsandevents/news/newsdetail.aspx?f=51719135 (accessed on 10 August 2026).
10. Kwak, Y.K.; Eberhard, M.O.; Kim, W.S.; Kim, J. Shear strength of steel fiber-reinforced concrete beams without stirrups. *ACI Struct. J.* **2002**, *99*, 530–538. [CrossRef] [PubMed]
11. Mansur, M.A.; Ong, K.C.G.; Paramasivam, P. Shear strength of fibrous concrete beams without stirrups. *J. Struct. Eng.* **1986**, *112*, 2066–2079. [CrossRef]
12. Al-Gburi, M.; Alhayani, A.A.; Almssad, A. Artificial neural network model for evaluating load capacity of RC deep beams. *Buildings* **2025**, *15*, 1371. [CrossRef]
13. Al-Gburi, M.; Jonasson, J.E.; Nilsson, M. Using artificial neural network to predict the restraint in concrete tunnel at early age. *Struct. Eng. Int.* **2015**, *25*, 258–265. [CrossRef]
14. Alhayani, A.A.; Almssad, A.; Al-Gburi, M. Size and location influence of opening on shear capacity of RC deep beams using ANN. *J. Asian Archit. Build. Eng.* **2026**, *25*, 1–15. [CrossRef]
15. Abambres, M.; Lantsoght, E.O. ANN-based shear capacity of steel fiber-reinforced concrete beams without stirrups. *Fibers* **2019**, *7*, 88. [CrossRef]
16. Yu, Y.; Zhao, X.Y.; Xu, J.J.; Wang, S.C.; Xie, T.Y. Evaluation of shear capacity of steel fiber reinforced concrete beams without stirrups using artificial intelligence models. *Materials* **2022**, *15*, 2407. [CrossRef] [PubMed]
17. Lantsoght, E.O.L. Database of shear experiments on steel fiber reinforced concrete beams without stirrups. *Materials* **2019**, *12*, 917. [CrossRef] [PubMed]
18. Hameed, A.A.; Al Sherrawi, M.H. Influence of steel fiber on the shear strength of a concrete beam. *Civ. Eng. J.* **2018**, *4*, 1699–1707. [CrossRef]
19. Bui, T.T.; Nana, W.S.A.; Doucet Ferru, B.; Bennani, A.; Lequay, H.; Limam, A. Shear performance of steel fiber reinforced concrete beams without stirrups: Experimental investigation. *Int. J. Civ. Eng.* **2020**, *18*, 897–912. [CrossRef]
20. Singh, B.; Jain, K. Appraisal of steel fibers as minimum shear reinforcement in concrete beams. *ACI Struct. J.* **2014**, *111*, 1191–1202. [CrossRef]
21. Moradi, M.; Aletaha, S.H.; Asadzadeh, E.; Mahmoodian Qomzade, M. Shear strength of fiber reinforced concrete beams with stirrups. *Iran. J. Sci. Technol. Trans. Civ. Eng.* **2021**, *46*, 865–882. [CrossRef]
22. Demakos, C.B.; Repapis, C.C.; Drivas, D.P. Experimental investigation of shear strength for steel fibre reinforced concrete beams. *Open Constr. Build. Technol. J.* **2021**, *15*, 81–90. [CrossRef]
23. Son, D.-H.; Chung, J.-H.; Choi, C.-S.; Bae, B.-I. Mitigation of size effect on shear capacity of steel fiber-reinforced concrete beams without stirrups. *Structures* **2026**, *85*, 111242. [CrossRef]
24. Zhao, M.; Li, J.; Li, C.; Shen, J. Effectiveness evaluation of different steel fibers on the shear strength of reinforced SFRC slender beams without web rebars. *Sci. Rep.* **2024**, *14*, 21234. [CrossRef] [PubMed]
25. Olden, J.D.; Jackson, D.A. Illuminating the "black box": A randomization approach for understanding variable contributions in artificial neural networks. *Ecol. Model.* **2002**, *154*, 135–150. [CrossRef]
26. Garson, G.D. Interpreting neural network connection weights. *AI Expert* **1991**, *6*, 47–51.
27. Pan, L.; Novák, L.; Lehký, D.; Novák, D.; Cao, M. Neural network ensemble-based sensitivity analysis in structural engineering: Comparison of selected methods and the influence of statistical correlation. *Comput. Struct.* **2021**, *242*, 106376. [CrossRef]
28. Kani, G.N.J. The riddle of shear failure and its solution. *ACI J. Proc.* **1964**, *61*, 441–468. Available online: https://www.concrete.org/publications/internationalconcreteabstractsportal.aspx?m=details&ID=7791 (accessed on 10 August 2026). [CrossRef] [PubMed]
29. Zararis, P.D.; Papadakis, G.C. Diagonal shear failure and size effect in RC beams without web reinforcement. *J. Struct. Eng.* **2001**, *127*, 733–742. [CrossRef]
30. Bazant, Z.P.; Kim, J.K. Size effect in shear failure of longitudinally reinforced beams. *ACI J.* **1984**, *81*, 456–468. Available online: http://www.civil.northwestern.edu/people/bazant/PDFs/Papers/172.pdf (accessed on 10 August 2026). [CrossRef] [PubMed]
31. Caterino, N.; Di Cristo, V.; Ceroni, F. Strength capacity of RC beams without shear reinforcement: Numerical analysis and comparisons with code provisions. *Eng. Struct.* **2024**, *317*, 118634. [CrossRef]

32. Jafari, A.; Shahmansouri, A.A.; Taslimi, A.; Naser, M.Z.; Zhou, Y. Shear capacity of FRP-RC beams: XML modeling and reliability evaluation. *Structures* **2025**, *81*, 110150. [CrossRef]
33. Ozkilic, Y.O.; Althaqafi, E.; Karalar, M.; Aksoylu, C. Shear and flexural performance of reinforced concrete beams with recycled concrete aggregates. *Rev. Adv. Mater. Sci.* **2025**, *64*, 20250140. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.